

WORKSHEET: DESCRIPTION OF MOTION

Chapter 14 | Mechanics Unit

Part I: Conceptual Understanding

1. Multiple Choice Questions (MCQs)

- a) A passenger sitting in a moving train is at rest relative to:
- (i) The station platform (ii) The train's seat (iii) A tree outside
- b) The path formed by the set of successive positions occupied by a moving object is the:
- (i) Displacement (ii) Abscissa (iii) Trajectory

2. True or False

- ☐ Motion and rest are absolute concepts. ()
- ☐ Displacement Δx can be negative if the mobile moves to the left. ()
- ☐ Instantaneous speed is measured over a very long time interval. ()

3. Completion

- A _____ is an object used for comparison to determine if others are moving.
- In 1D motion, the position of a mobile M is determined by its _____ x .
- The SI unit for acceleration is _____.

Part II: Vectors and Representation

4. The Characteristics of Velocity List the four elements required to represent an instantaneous velocity vector $\vec{v}$:

1. _____
2. _____
3. _____
4. _____

Part III: Numerical Calculations

5. Speed and Duration A mobile travels from $x_i = -10$ m to $x_f = 50$ m in a duration of 4 seconds.

- i. Calculate the displacement Δx : _____
- ii. Calculate the average speed v_{av} in m/s: _____
- iii. Convert the speed to km/h: _____

6. Nature of Motion A car's velocity changes from $v_1 = 20 \text{ m/s}$ to $v_2 = 5 \text{ m/s}$ in 3 seconds.

- i. Calculate the average acceleration a_{av} : _____
- ii. Determine the nature of motion (Accelerated or Decelerated). Justify using the signs of v and a :

Part IV: Chronophotography (Advanced)

7. Formula Application & Acceleration Deduction A recording shows a sequence of points M_1, M_2, M_3, M_4, M_5 where the constant time interval is $\tau = 0.04 \text{ s}$. The following distances are measured: $M_1M_3 = 8 \text{ cm}$ and $M_3M_5 = 12 \text{ cm}$.

1. Calculate the instantaneous speed v_2 at point M_2 in m/s:

2. Calculate the instantaneous speed v_4 at point M_4 in m/s:

3. Calculate the instantaneous acceleration a_3 at point M_3 in m/s^2 :

4. Based on the change in velocity, specify the nature of motion between M_2 and M_4 .

Teacher's Note: Show all calculation steps. Pay close attention to SI units (m, s, m/s^2). Note that $1 \text{ cm} = 0.01 \text{ m}$.